ObelyZK: Verifiable Linear-Trace Inference for Billion-Parameter Neural Networks via GKR Sumcheck with Aggregated Weight Binding on Starknet

Victor Amaya, Bitsage Network

info@obelysk.xyz

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Abstract. We present ObelyZK, a system that cryptographically verifies the *linear arithmetic trace* of neural-network inference—matrix multiplications, additions, and scaling operations—for models up to 14 billion parameters, deployed on Starknet. The core contribution is *aggregated weight binding*: a mismatch-sumcheck protocol that reduces the on-chain calldata required to verify 160 weight matrices from 2.4 M felts to 86,723 felts (≈28×), making streaming verification of LLM-scale proofs practical within Starknet's per-transaction step limits. The system operates over the Mersenne prime field $p = 2^{31} - 1$, extended through a degree-4 tower M31 → CM31 → QM31 providing 124-bit algebraic security, and achieves 2.58 s amortized per-layer GPU proving on NVIDIA H100 (103 s total for the full 40-layer model). The on-chain Cairo verifier (∼14 K lines) accepts proofs via 18 streaming transactions and is deployed on Starknet Sepolia. We provide formal soundness analysis: at the recommended operational setting of $q = 10$ MLE spot-check queries, the end-to-end forgery probability for the linear trace is bounded by 2^{-111} (conservatively; the MLE opening component alone achieves 2^{-130}). **Scope:** The deployed verifier (v34) constrains linear operations, nonlinear activations via range-reduced LogUp (GELU, Sigmoid, Softmax verified on lower 16–20 bits; ReLU via algebraic proof), KV-cache commitment chains with on-chain decode chain validation (RoPE position offset binding, cross-proof continuity enforcement), and the full Fiat-Shamir transcript via a 10-round-hardened replay verifier covering all 12 GKR layer tags, deferred proofs, and weight binding. LayerNorm/RMSNorm mean/variance remain unconstrained; we discuss the path to full inference verification.

Keywords: verifiable inference, linear arithmetic trace, GKR protocol, sumcheck, aggregated weight binding, STARK, Starknet, M31

1 Introduction

Verifying that a specific neural network with committed weights produced a specific output for a given input is a fundamental problem for AI accountability, regulatory compliance, and trustless computation markets [13, 22]. Existing ZKML systems face a scalability wall: EZKL [6] handles ∼1 M parameters via Halo2/KZG; Giza [8] uses Cairo-STARK for small models; zkLLM [22] demonstrates 13 B-parameter proving but requires 1–15 minutes and lacks a deployed on-chain verifier.

The central obstacle at LLM scale is *on-chain calldata*: a 40-layer transformer with 160 weight matrices requires ∼2.4 M field elements for individual MLE openings, exceeding any L2's per-transaction limits. This paper's main contribution addresses this bottleneck directly.

1.1 Contributions

1. **Aggregated weight binding** (Section 5) — a mismatch-sumcheck protocol that batches 160 weight-matrix MLE openings into a single global oracle, reducing calldata from 2.4 M to 86,723 felts (≈28×). This is the key enabler for on-chain LLM verification.
2. **Cairo-native GKR streaming verifier** (Section 7) — ∼14 K lines of Cairo implementing the complete field tower, Fiat-Shamir channel, sumcheck, MLE opening, and aggregated binding verification across 18 streaming transactions, including KV-cache commitment verification for decode-step chaining. Deployed on Starknet Sepolia.
3. **GPU-accelerated GKR proving** (Section 6) — 103 s end-to-end for the full 40-layer Qwen3-14B on NVIDIA H100 NVL (2.58 s amortized per layer), with 17+ custom CUDA kernels.
4. **Incremental KV-cache proving** (Section 3.3) — $O(\log T)$ per-step commitment via incremental Poseidon Merkle trees, with on-chain decode chain enforcement (RoPE position offset binding, cross-proof continuity validation), and weight commitment caching.
5. **Formal security analysis** (Section 8) — four theorems establishing soundness bounds for each verification component, hardened over 10 rounds of review with $q = 10$ MLE queries (130-bit opening security).

1.2 Scope and Honest Assessment

Table 1 summarizes what is deployed, implemented, and specified. The deployed verifier (v34) cryptographically constrains *linear operations* (matmul, addition, scaling), *nonlinear activations* via range-reduced LogUp [11] (enforced by default), *KV-cache commitment chains* with decode chain enforcement (position offset binding and cross-proof continuity), and *the full Fiat-Shamir transcript* via a 10-round-hardened replay verifier covering all 12 GKR layer tags, deferred proofs, IO commitment, and weight binding with $q = 10$ MLE queries (130-bit opening security). Activation inputs are masked to table range (2^{16}–2^{20} entries) before proving; see Section 10 for the remaining gap (LayerNorm/RMSNorm mean/variance).

Table 1: Implementation status matrix. ✓ = yes, ✗ = no, ∘ = partial.

Component	Spec	Impl	Depl	Bench
MatMul sumcheck	✓	✓	✓	✓
Aggregated binding	✓	✓	✓	✓
Streaming verifier	✓	✓	✓	✓
MLE opening	✓	✓	✓	✓
Weight cache	✓	✓	✓	✓
Activation LogUp	✓	✓	✓	∘
KV-cache decode	✓	✓	✓	✓
Fiat-Shamir replay	✓	✓	✓	—
Privacy layer	✓	✓	✓	∘

Spec = protocol specified; Impl = code implemented; Depl = deployed on Starknet Sepolia; Bench = included in reported benchmarks.

1.3 Paper Organization

Section 2 presents the M31 field tower and MLE foundations. Sections 3 and 4 detail the GKR and sumcheck protocols. Section 5 presents aggregated weight binding (the core contribution). Section 6 discusses GPU acceleration. Section 7 details on-chain streaming verification. Section 8 provides a theorem-driven security analysis. Section 9 reports benchmarks with a comparison to prior work. Section 10 discusses limitations and future work. Section 11 surveys related work. Section 12 concludes.

2 Preliminaries and Field Tower

2.1 The Mersenne Prime Field M31

All arithmetic operates over $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ with $p = 2^{31} - 1 = 2{,}147{,}483{,}647$.

Lemma 2.1 (Efficient reduction)**.** *For $a, b \in \mathbb{F}_p$, let $w = a \cdot b$ as a 62-bit integer. Write $w = lo + hi \cdot 2^{31}$ where $lo = w \bmod 2^{31}$ and $hi = \lfloor w/2^{31} \rfloor$. Then*

$$a \cdot b \bmod p = \begin{cases} lo + hi & \text{if } lo + hi < p, \\ lo + hi - p & \text{otherwise.} \end{cases} \tag{1}$$

Proof. Since $p = 2^{31} - 1$, we have $2^{31} \equiv 1 \pmod p$. Therefore $w = lo + hi \cdot 2^{31} \equiv lo + hi \pmod p$. Now $lo \in [0, 2^{31})$ and $hi \in [0, 2^{31})$ (since $w < p^2 < 2^{62}$), so $lo + hi \in [0, 2^{32} - 2]$. Since $p = 2^{31} - 1$, the sum can be at most $2p$, so at most one subtraction of p suffices to reduce to $[0, p)$. □

Field operations map directly to SIMD/GPU instructions:

- **Addition:** $a + b$; subtract p if $\geq p$.
- **Multiplication:** 64-bit multiply, shift, mask, add, conditional subtract (Theorem 2.1).
- **Inversion:** $a^{p-2} \bmod p$ via Fermat's little theorem, using an addition chain of length 37.

2.2 Field Extension Tower

Definition 2.2 (CM31 — Complex Extension)**.** CM31 = $\mathrm{M31}[i]/(i^2 + 1)$, elements $z = a + bi$ with $a, b \in \mathrm{M31}$. The polynomial $x^2 + 1$ is irreducible since -1 is a quadratic non-residue mod p (because $p \equiv 3 \pmod 4$).

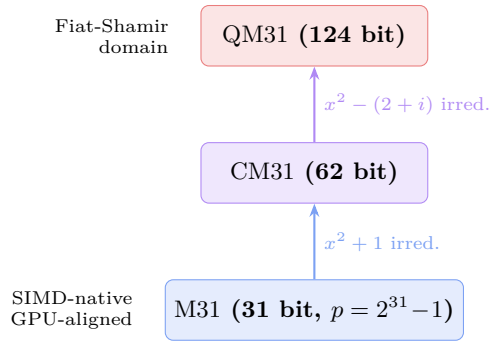

Figure 1: M31 field extension tower. The base field enables hardware-efficient arithmetic; the quartic extension provides 124-bit security for the Fiat-Shamir channel.

CM31 multiplication: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$.

Definition 2.3 (QM31 — the Secure Field)**.** QM31 = $\mathrm{CM31}[j]/(j^2 - (2+i))$, elements $\alpha + \beta j$ with $\alpha, \beta \in \mathrm{CM31}$. Each QM31 element has $4 \times 31 = 124$ bits of algebraic security.

Lemma 2.4 (QM31 Karatsuba multiplication)**.** *For $x = \alpha_1 + \beta_1 j$ and $y = \alpha_2 + \beta_2 j$ in* QM31*:*

$$x \cdot y = \big(\alpha_1\alpha_2 + \beta_1\beta_2 \cdot u\big) + \big((\alpha_1+\beta_1)(\alpha_2+\beta_2) - \alpha_1\alpha_2 - \beta_1\beta_2\big)j, \tag{2}$$

where $u = 2 + i$ is the irreducible element and the multiplication by u in CM31 *is: $(a+bi)\cdot(2+i) = (2a-b)+(a+2b)i$.*

Proof. Expanding $(\alpha_1 + \beta_1 j)(\alpha_2 + \beta_2 j)$ and using $j^2 = 2 + i = u$ gives the real part $\alpha_1\alpha_2 + \beta_1\beta_2\, u$ and cross term $\alpha_1\beta_2 + \beta_1\alpha_2$. The Karatsuba identity $\alpha_1\beta_2 + \beta_1\alpha_2 = (\alpha_1+\beta_1)(\alpha_2+\beta_2) - \alpha_1\alpha_2 - \beta_1\beta_2$ reduces from 4 to 3 CM31 multiplications. □

2.3 Multilinear Extensions (MLEs)

Definition 2.5 (Multilinear extension)**.** For $f : \{0,1\}^n \to \mathbb{F}$, the unique multilinear polynomial agreeing with f on the Boolean hypercube is:

$$\widetilde{f}(x_1, \ldots, x_n) = \sum_{y \in \{0,1\}^n} f(y) \prod_{i=1}^{n} \big(x_i y_i + (1-x_i)(1-y_i)\big). \tag{3}$$

The *Lagrange equality kernel* is $\mathrm{eq}(\mathbf{x}, \mathbf{y}) = \prod_{i=1}^{n}(x_i y_i + (1 - x_i)(1 - y_i))$, evaluating to 1 when $\mathbf{x} = \mathbf{y}$ on the Boolean hypercube.

MLE folding. Fixing variable $x_1 = r$ reduces the MLE dimension by one:

$$\widetilde{f}|_{x_1=r}(x_2, \ldots, x_n) = (1-r) \cdot \widetilde{f}(0, x_2, \ldots) + r \cdot \widetilde{f}(1, x_2, \ldots). \tag{4}$$

This operation is linear in the table size and is the core primitive for both the sumcheck prover and the MLE opening verifier.

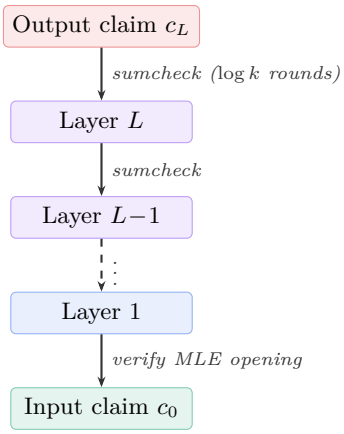

Figure 2: GKR claim reduction: each layer reduces an output claim to an input claim via sumcheck. After L reductions, the verifier checks the final claim against the committed input MLE.

3 GKR for Neural Networks

3.1 Protocol Overview

The GKR protocol [9, 23] verifies layered arithmetic circuits by reducing an output claim to an input claim one layer at a time. For a neural network with L layers:

$$\text{Input} \xrightarrow{\text{Layer 1}} h_1 \xrightarrow{\text{Layer 2}} h_2 \xrightarrow{\cdots} h_{L-1} \xrightarrow{\text{Layer } L} \text{Output}.$$

The verifier starts with a claim about the output and reduces it backward through each layer via the sumcheck protocol [17]. Per-layer verification cost is $O(\log n)$ for layer width n, regardless of the actual computation size.

3.2 Layer Types

The GKR circuit supports 12 layer tags (Table 2), including quantization/dequantization layers with metadata-keyed LogUp, embedding layers with sparse-table lookups, and the `AttentionDecode` tag for incremental KV-cache decode.

Self-attention with Grouped Query Attention (GQA) computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

We decompose this into five sublayers: QK^T matmul (degree 2), scaling (degree 1), causal mask (linear), softmax (range-reduced LogUp), and score×V matmul (degree 2). RoPE encoding is a linear layer with position-dependent trigonometric coefficients precomputed as an MLE.

3.3 KV-Cache Decode-Step Proving

Autoregressive generation produces one token per decode step. Naïvely re-proving the entire sequence each step is $O(T \cdot L)$; instead, we cache K/V projections and prove only the new token's computation.

Commitment chain. Each decode step produces a GKR proof binding two Poseidon commitments: C_{prev} (cache state before the step) and C_{new} (after appending the new K/V). Both are mixed into the Fiat-Shamir channel before the circuit metadata, allowing sequential chaining: step N's C_{prev} must equal step $(N-1)$'s C_{new}.

Incremental Merkle commitment. Computing C_{new} via flat hashing costs $O(\text{cache_size})$ Poseidon calls per step—524,288 hashes at $T = 4096$ for a 40-layer model. We replace this with an incremental binary Merkle tree per layer: each position's K/V data across all heads is hashed into a single leaf, and `push()` recomputes only the $O(\log T)$ path to root. The model-level commitment hashes sorted layer roots: $C = \text{Poseidon}_{\text{many}}([\texttt{0x4B56}, L, id_0, r_0, id_1, r_1, \ldots])$. At $T = 4096$ this is ∼140 hashes per step (∼3,700× reduction). The tree supports automatic capacity doubling when the sequence exceeds the initial allocation.

Weight commitment cache. Decode steps re-use the same weight matrices. A `SharedWeightCache` stores Poseidon Merkle roots keyed by $(node_id, rows, cols)$; the first step populates the cache, and subsequent steps skip the entire commitment pipeline (0.8 s → <1 ms per step for the benchmark model).

RoPE position offset binding. The `AttentionDecode` proof carries a position_offset = full_seq_len − new_tokens field encoding the RoPE rotation base for the new query. The prover computes and mixes this into the Fiat-Shamir channel after the `causal` flag; the verifier replays the mix and asserts the invariant position_offset + new_tokens = full_seq_len. The Cairo on-chain verifier additionally enforces: new_tokens > 0, full_seq_len ≥ new_tokens, d_{model} mod num_heads = 0, and num_sub_proofs = 4 + 2 · num_heads.

On-chain decode chain registry. The Cairo contract maintains a per-model decode chain: (last_kv_commitment, last_full_seq_len, initialized). When the `validate_decode_chain` flag is set in `stream_init`, the finalize step asserts: (a) $C_{\text{prev}} = C_{\text{chain}}$ (previous KV commitment matches the chain's last recorded commitment), and (b) position_offset = last_full_seq_len (the new proof continues exactly where the last left off). After verification, the chain registry is updated with the current KV commitment and sequence length. The `ModelGkrVerified` event emits (position_offset, full_seq_len, new_tokens) for external auditing.

3.4 Batch Verification

Multiple layer instances $\{I_1, \ldots, I_m\}$ with claims $\{c_1, \ldots, c_m\}$ are batched via random linear combination. The verifier draws $\alpha, \lambda \leftarrow \text{QM31}$ from the Fiat-Shamir channel and forms the batched claim:

$$S = \sum_{i=1}^{m} \alpha^i \cdot 2^{n_{\max} - n_i} \cdot \left(\sum_j \lambda^j \cdot c_{i,j}\right), \tag{5}$$

where $n_{\max} - n_i$ is the padding doubling factor for instance i.

Algorithm 1 Sumcheck round for MatMul verification

Require: MLEs f_a, f_b of length $2 \cdot \text{mid}$; current claim
1: $S_0 \leftarrow \sum_{i=0}^{\text{mid}-1} f_a[i] \cdot f_b[i]$
2: $S_1 \leftarrow \sum_{i=0}^{\text{mid}-1} f_a[\text{mid}+i] \cdot f_b[\text{mid}+i]$
3: $S_2 \leftarrow \sum_{i=0}^{\text{mid}-1} (2f_a[\text{mid}+i] - f_a[i])(2f_b[\text{mid}+i] - f_b[i])$
4: Lagrange interpolate: $p(X) = c_0 + c_1 X + c_2 X^2$
5: **assert** $c_0 + (c_0 + c_1 + c_2) = \text{claim}$ $\triangleright\ p(0) + p(1)$
6: Send (c_0, c_2); verifier reconstructs $c_1 = \text{claim} - 2c_0 - c_2$
7: $r_t \leftarrow$ Fiat-Shamir challenge
8: Fold: $f_a^{(t+1)}[i] = f_a[i] + r_t\,(f_a[\text{mid}+i] - f_a[i])$
9: Update claim: $p(r_t) = c_0 + r_t(c_1 + r_t\, c_2)$ (Horner)

3.5 Fiat-Shamir Channel

The interactive protocol is made non-interactive via a Poseidon252-based Fiat-Shamir channel [7]:

- State: (digest : felt252, n_draws : u32).
- `mix_felt(v)`: digest $\leftarrow$ Hades(digest, v, 2)[0]; $n_\text{draws} \leftarrow 0$.
- `draw_qm31()`: draw felt252, extract 4 M31 components via successive mod 2^{31}.

4 Sumcheck for Matrix Multiplication

4.1 Problem Formulation

Given $A \in \mathbb{F}^{m\times k}$, $B \in \mathbb{F}^{k\times n}$, claimed $C = AB$, and random evaluation points $\mathbf{r}_i \in \mathbb{F}^{\log m}$, $\mathbf{r}_j \in \mathbb{F}^{\log n}$:

$$\widetilde{C}(\mathbf{r}_i, \mathbf{r}_j) = \sum_{\mathbf{x}\in\{0,1\}^{\log k}} \widetilde{A}(\mathbf{r}_i, \mathbf{x}) \cdot \widetilde{B}(\mathbf{x}, \mathbf{r}_j). \tag{6}$$

This is a sum over k terms of a degree-2 polynomial (product of two MLEs), exactly the form required by the sumcheck protocol.

MLE layout. Matrix A uses row-major layout; matrix B uses column-major (transposed) layout, enabling fixing the k-dimension variables first.

4.2 Round Protocol

Proof compression. Only (c_0, c_2) are transmitted per round; the verifier reconstructs $c_1 = \text{claim} - 2c_0 - c_2$, saving 33% of calldata.

5 Aggregated Weight Binding

This section presents the paper's core contribution: a protocol that makes on-chain verification of LLM-scale weight commitments practical.

5.1 The Calldata Bottleneck

A 40-layer Qwen3-14B has 160 weight matrices. Verifying each independently requires an MLE opening proof per matrix, totaling ∼2.4 M felts of calldata. Starknet's per-transaction step limit (∼10 M Cairo steps) makes processing this volume infeasible in streaming transactions.

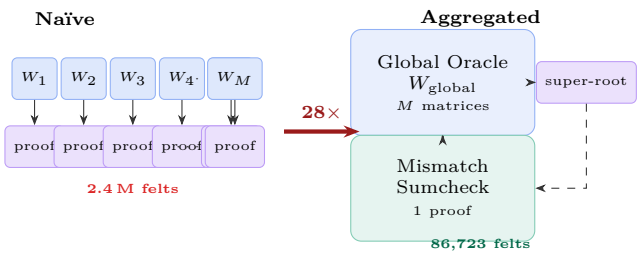

Figure 3: Aggregated weight binding eliminates per-matrix opening proofs. Left: naïve approach requires one MLE opening per weight matrix (∼2.4 M felts). Right: a single global oracle with mismatch sumcheck verifies all 160 matrices (86,723 felts), a 28× reduction.

5.2 Global Oracle Construction

Definition 5.1 (Global weight oracle)**.** Given M weight matrices $\{W_1, \ldots, W_M\}$ with dimensions $\{(r_i, c_i)\}_{i=1}^M$, the global oracle MLE is:

$$W_{\text{global}}(\mathbf{s}, \mathbf{x}) = \sum_{i=1}^{M} \text{eq}(\mathbf{s}, \mathbf{g}_i) \cdot \widetilde{W_i}(\mathbf{x}), \tag{7}$$

where $\mathbf{s} \in \{0,1\}^{\lceil \log_2 M \rceil}$ are selector bits and $\mathbf{g}_i$ is the binary encoding of matrix index i.

5.3 Mismatch Sumcheck Protocol

Protocol 5.2 (Aggregated weight binding)**.** Given M weight matrices with claimed evaluations v_i at points $\mathbf{e}_i$ (produced by the GKR sumcheck):

1. **Batch.** Draw $\rho \leftarrow$ QM31 from the Fiat-Shamir channel.
2. **Mismatch sumcheck.** Prove that the random linear combination of all mismatches vanishes:

$$\sum_{i=1}^{M} \rho^i \cdot \text{eq}(\mathbf{g}_i, \mathbf{s}) \cdot \big(W_{\text{global}}(\mathbf{s}, \mathbf{e}_i) - v_i\big) \equiv 0. \tag{8}$$

3. **Commitment check.** Verify the final MLE opening against the committed Poseidon2-M31 super-root with subtree Merkle openings.

Intuition. If all claimed evaluations are correct ($W_{\text{global}}(\mathbf{s}, \mathbf{e}_i) = v_i$ for all i), every term in the sum is zero. A cheating prover who forges even one evaluation v_j must produce a non-zero polynomial that sums to zero—which by Schwartz-Zippel [20, 26] occurs with probability at most $1/|\text{QM31}| \approx 2^{-124}$.

5.4 Zero-Tree Commitment Extension

Matrices with different depths are padded to $n_{\max}$ using zero-tree hashing: $h_0 = \text{pack}(\text{QM31} :: 0)$, $h_i = \text{poseidon_hash}(h_{i-1}, h_{i-1})$. This allows a single Merkle super-root to cover all weight matrices regardless of individual dimensions.

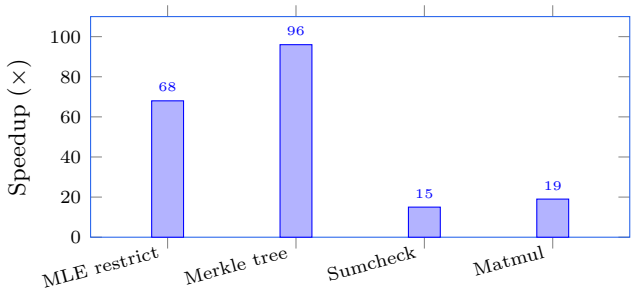

Figure 4: GPU (CUDA) speedup over CPU (STWO SIMD) for key operations on NVIDIA H100.

5.5 Streaming Calldata Breakdown

5.6 Memory-Efficient Proving

The off-chain binding prover uses `GraphWeightSource`—a streaming interface that loads weight matrices on demand from the model graph rather than materializing all 160 matrices simultaneously. This bounds peak memory at ∼4 GB (one weight matrix at a time) versus ∼640 GB for full materialization.

5.7 Weight Commitment Cache

Poseidon2-M31 Merkle root computation for 160 matrices takes ∼500 s on first run. The weight cache (`.swcf` file) stores roots keyed by $(node_id, rows_padded, cols_padded)$, reducing subsequent proofs to <1 ms (cache hit) for the commitment phase.

6 GPU-Accelerated Proving

The proving pipeline employs 17+ custom CUDA kernels totaling 20 K+ lines, achieving 68–96× speedup over STWO's SIMD backend for ML-specific workloads.

Sumcheck kernel. Block size 256, shared memory 12 KB/block, dispatch threshold $k \geq 2^{14}$. Per-thread: load $(f_a[i], f_a[\text{mid}+i], f_b[i], f_b[\text{mid}+i])$, compute partial S_0, S_1, S_2, then 8-iteration binary tree reduction with `__syncthreads()`.

MLE fold kernel. 512 threads/block, grid $= \lceil n/(2 \cdot 512) \rceil$. Output $[i] = f[i] + r \cdot (f[\text{mid}+i] - f[i])$ with challenge broadcast (16 bytes).

GPU M31 arithmetic.

$$\begin{aligned} \texttt{m31_mul}(a,b) : \text{prod} &= (\text{u64})\, a \cdot b; \\ \text{lo} &= \text{prod} \,\&\, \texttt{0x7FFFFFFF}; \\ \text{hi} &= \text{prod} \gg 31; \\ &\textbf{return}\ \texttt{m31_add}(\text{lo}, \text{hi}). \end{aligned}$$

Multi-GPU support. Weight commitment computation supports multi-GPU via greedy bin-packing (`multi_gpu::partition_by_size()`), with one `DeviceGuard` per thread and fallback to single-GPU pipelined path on failure.

7 On-Chain Streaming Verification

7.1 Cairo Verifier Architecture

The on-chain verifier (v34) comprises ∼14 K lines of Cairo across 21 modules. A Rust-side *replay verifier* independently re-derives the Fiat-Shamir channel state from serialized proof data before submission, catching prover/serializer mismatches pre-flight (Theorem 8.4, item (f)). Class hash:

`0x6a6b...d5ec1f63d715...ffbaf6c5b5439`.

7.2 MLE Opening Verification

1. Commitment: 2^n evaluations → Poseidon2-M31 Merkle root.
2. Folding: n rounds, challenge r_i per round: $f_{i+1}[j] = f_i[j] + r_i(f_i[\text{mid}+j] - f_i[j])$.
3. Spot checks: q queries per fold (read from proof at runtime; default 5 via `MLE_NUM_QUERIES`, recommended $q = 10$ for production; see Theorem 8.2).
4. Verify Merkle paths and folding consistency at each layer.

7.3 18-Transaction Streaming Protocol

The streaming protocol (Figure 5) splits verification across 18 transactions to fit within Starknet's per-TX step limit:

1. **Session phase** (TXs 1–4): open session, upload calldata in chunks, seal session.
2. **Verification phase** (TXs 5–18): init channel state, verify output MLE, process 10 layer steps (each running sumcheck verification), verify aggregated weight binding, finalize with input MLE check and verdict.

QM31 packing. QM31 values are packed into felt252: packed $= 1 \cdot 2^{124} + a_0 \cdot 2^{93} + a_1 \cdot 2^{62} + b_0 \cdot 2^{31} + b_1$ (125 bits used, 1-bit sentinel prevents zero ambiguity).

IO-packed calldata. 8 M31 per felt252, reducing IO calldata from 10,246 to 1,281 felts. `evaluate_mle_from_packed_1row()` evaluates the MLE directly from packed felts without unpacking.

KV-cache commitment verification. For decode-step proofs, the Cairo verifier accepts two additional felt252 values (`kv_cache_commitment` and `prev_kv_cache_commitment`) in the streaming init transaction, along with decode metadata (`position_offset`, `full_seq_len`, `new_tokens`, `validate_decode_chain`). KV commitments are mixed into the Fiat-Shamir channel before `circuit_depth`, and all fields are included in the proof hash and `ModelGkrVerified` event.

Decode chain enforcement. When `validate_decode_chain` is set, the finalize transaction reads the per-model decode chain registry and asserts sequential continuity: the previous KV commitment must match the chain's last value, and position_offset must equal the chain's last full_seq_len. This prevents a malicious prover from skipping intermediate decode steps or substituting a different KV-cache state. Three view functions (`get_decode_chain_kv_commitment`,

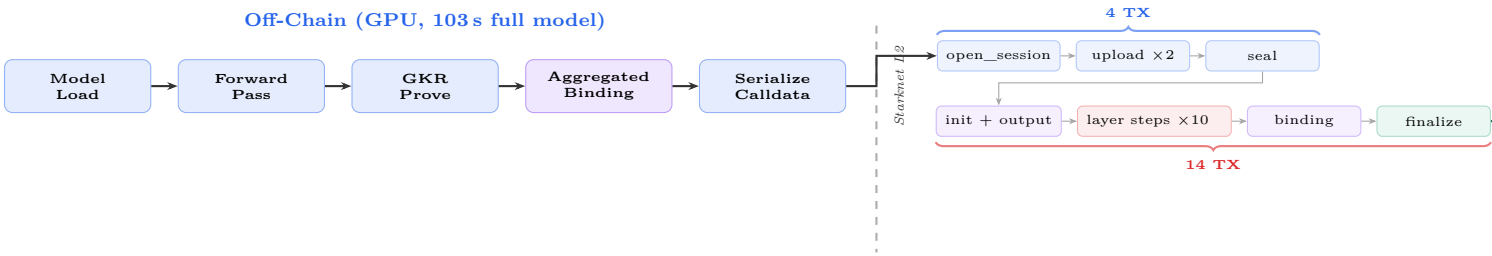

Figure 5: End-to-end proof flow. Off-chain: model loading, M31 forward pass, GKR proving, aggregated binding, and calldata serialization. On-chain: 18 Starknet transactions split into 4 session management and 14 streaming verification steps, ending with an accept/reject verdict.

`get_decode_chain_full_seq_len`, `is_decode_chain_initialized`) expose chain state for external monitoring.

8 Security Analysis

We present four theorems establishing the soundness of each verification component. All security statements are in the random oracle model for Fiat-Shamir.

8.1 Threat Model

We consider a computationally unbounded malicious prover $\mathcal{P}^*$ attempting to convince the on-chain verifier $\mathcal{V}$ to accept an incorrect result. Assumptions:

- The Fiat-Shamir hash (Poseidon252) is modeled as a random oracle binding transcript challenges to the full history.
- Starknet L2 provides data availability and transaction ordering guarantees.
- The prover must produce a valid transcript accepted by the Cairo verifier contract; no physical access to the verifier is assumed.

8.2 Theorem 1: Sumcheck Soundness Per Layer

Theorem 8.1 (Sumcheck soundness)**.** *For a degree-d sumcheck polynomial over* QM31 *with k rounds, a cheating prover passes with probability at most $kd/|\mathrm{QM31}|$ [17, 20, 26]. For matmul with $k = \lceil \log_2 5120 \rceil = 13$ and $d = 2$:*

$$\varepsilon_{sc} \leq \frac{13 \times 2}{2^{124}} = \frac{26}{2^{124}} \approx 2^{-119}.$$

Proof. Each sumcheck round requires the prover to send a degree-d univariate polynomial $p_i(X)$ such that $p_i(0) + p_i(1)$ equals the current claim. By Schwartz-Zippel, a dishonest p_i agrees with the honest polynomial at the random challenge r_i with probability at most $d/|\mathrm{QM31}|$. By union bound over k rounds, the total cheating probability is at most $kd/|\mathrm{QM31}|$. □

8.3 Theorem 2: MLE Opening Soundness

Theorem 8.2 (MLE opening soundness)**.** *Consider the MLE opening protocol with n folds and q spot-check queries per fold, where queries are sampled uniformly without replacement from the folded table at each round. At fold i, the table has 2^{n-i} entries and q positions are checked. A cheating prover who substitutes an incorrect folded value at any position must avoid all q checks at that fold. Under the assumption that the Fiat-Shamir transcript (which determines query positions) behaves as a random oracle, the probability of evading detection at a single fold is:*

$$\Pr[\textit{evade fold } i] \leq 1 - \frac{q}{2^{n-i}}.$$

Since $2^{n-i} \geq 2$ for all folds ($i < n$), this gives $\Pr[\textit{evade}] \leq 1 - q/2^{n-i} \leq (1 - 1/2)^q = 2^{-q}$ as a conservative per-fold bound (worst case at the final fold where the table has only 2 entries).

The total evasion probability over n folds is then conservatively bounded by:

$$\varepsilon_{mle} \leq 2^{-qn}. \tag{9}$$

For $n = 13$ folds:

- *$q = 5$ (current default): $\varepsilon_{mle} = 2^{-65}$.*
- *$q = 10$ (recommended): $\varepsilon_{mle} = 2^{-130}$.*

This is a conservative upper bound under the random oracle model; the true soundness error is strictly smaller at earlier folds.

Proof. At fold i, the verifier draws q query indices uniformly from $\{0, \ldots, 2^{n-i} - 1\}$ (without replacement), then checks that each queried position is consistent with the committed Merkle path and the folding equation $f_{i+1}[j] = f_i[j] + r_i(f_i[\mathrm{mid} + j] - f_i[j])$.

If the prover corrupts any entry in the folded table, the probability that none of the q queries lands on a corrupted position is at most $(1 - 1/2^{n-i})^q$. At the last non-trivial fold ($i = n - 1$, table size 2), this is $(1/2)^q = 2^{-q}$, the worst case.

Conditioning across folds. The adversary must commit to the Merkle root of each folded table *before* the Fiat-Shamir oracle reveals that fold's query positions. Under the random oracle model, these query positions are independent of the adversary's committed table. We condition on each fold sequentially: let E_i be the event that the adversary evades detection at fold i. Then $\Pr[E_0 \cap \cdots \cap E_{n-1}] = \prod_i \Pr[E_i \mid E_0 \cap \cdots \cap E_{i-1}]$. Since the adversary commits fold i's table before seeing fold i's

queries, $\Pr[E_i \mid \cdot] \le 2^{-q}$ regardless of prior events. Thus $\varepsilon_{\text{mle}} \le \prod_{i=0}^{n-1} 2^{-q} = 2^{-qn}$.

Note: This uses the worst-case per-fold bound $(1/2)^q$ from the final fold (table size 2). At earlier folds (larger tables), evasion probability is strictly less than 2^{-q}, so the true error is smaller than 2^{-qn}. We use 2^{-qn} as a clean, conservative bound. □

Operational default. The intended production configuration is $q = 10$, providing 130-bit MLE opening security that matches the field's 124-bit algebraic security margin. The current Cairo default of $q = 5$ is a development convenience; all soundness bounds in this paper are stated parametrically in q so that either setting can be evaluated. The calldata overhead of $q = 10$ vs. $q = 5$ is 2× Merkle paths per fold, adding ∼4,000 felts to the 86,723 total—a 4.6% increase for 65 additional bits of security. The query count is a runtime parameter read from the proof structure; changing q requires no code modifications.

8.4 Theorem 3: Aggregated Weight Binding Soundness

Theorem 8.3 (Aggregated binding soundness)**.** *Consider M weight matrices with GKR-produced claimed evaluations $\{v_i\}_{i=1}^M$ at points $\{\mathbf{e}_i\}$, and suppose at least one claim is forged: $v_j \neq W_{global}(\mathbf{g}_j, \mathbf{e}_j)$ for some j. The mismatch sumcheck (Theorem 5.2) accepts with probability at most:*

$$\varepsilon_{bind} \le \frac{M + k_{sc} \cdot d_{sc}}{|\mathrm{QM31}|} \tag{10}$$

where k_{sc} is the number of sumcheck rounds (equal to the number of variables in the global oracle, $\lceil \log_2 M \rceil + \max_i \lceil \log_2 |W_i| \rceil$, since the sumcheck ranges over the full selector + data dimensions of W_{global}) and $d_{sc} = 2$ is the per-round polynomial degree (the product $\mathrm{eq}(\mathbf{g}_i, \mathbf{s}) \cdot W_{global}(\mathbf{s}, \mathbf{e}_i)$ is degree 1 in each variable of $\mathbf{s}$, and the batching multiplier ρ^i is a constant with respect to $\mathbf{s}$, giving total degree 2 per round after the mismatch term). For $M = 160$ and $k_{sc} = \lceil \log_2 N_{global} \rceil \le 28$:

$$\varepsilon_{bind} \le \frac{160 + 56}{2^{124}} = \frac{216}{2^{124}} < 2^{-116}.$$

Proof. Define the mismatch polynomial:

$$P(\rho, \mathbf{s}) = \sum_{i=1}^{M} \rho^i \cdot \mathrm{eq}(\mathbf{g}_i, \mathbf{s}) \cdot \big(W_{\text{global}}(\mathbf{s}, \mathbf{e}_i) - v_i\big).$$

If any v_j is forged, the term $\rho^j \cdot \mathrm{eq}(\mathbf{g}_j, \mathbf{s}) \cdot \delta_j$ is non-zero (where $\delta_j = W_{\text{global}}(\mathbf{g}_j, \mathbf{e}_j) - v_j \neq 0$). Since P has degree at most M in ρ and the terms ρ^j are linearly independent, P is a non-zero polynomial in ρ.

The protocol proceeds in two phases:

1. **Batching:** The verifier draws $\rho \leftarrow \mathrm{QM31}$ via Fiat-Shamir. By Schwartz-Zippel, the probability that $P(\rho, \mathbf{s})$ is identically zero in $\mathbf{s}$ (for this choice of ρ) is at most $M/|\mathrm{QM31}|$.
2. **Sumcheck:** Conditioned on $P(\rho, \cdot) \not\equiv 0$, the sumcheck over $\mathbf{s}$ has k_{sc} rounds of degree $d_{\text{sc}} = 2$. By Theorem 8.1's argument, cheating probability is at most $k_{\text{sc}} d_{\text{sc}}/|\mathrm{QM31}|$.

By union bound, the total probability is $(M + k_{\text{sc}} d_{\text{sc}})/|\mathrm{QM31}|$.

The final Merkle commitment check (verifying the reduced MLE evaluation against the Poseidon2-M31 superroot) is deterministic and adds no soundness error beyond the hash collision bound ($\sim 2^{-124}$ for Poseidon2-M31), which is absorbed into the field-size denominator. □

8.5 Theorem 4: Deployed System Soundness

Theorem 8.4 (End-to-end deployed soundness)**.** *The deployed v33 verifier constrains the following:*

(a) Every linear layer's sumcheck reduction is consistent (matmul, addition, scaling, mul).
(b) All M weight MLE evaluations match committed Poseidon2-M31 roots via aggregated binding.
(c) MLE opening spot-checks pass at each fold.
(d) The Fiat-Shamir transcript is honestly constructed.
(e) Nonlinear activations (GELU, Sigmoid, Softmax) match the precomputed table on the lower 2^{16}–2^{20} bits via range-reduced LogUp; ReLU is verified algebraically.
(f) A Rust-side Fiat-Shamir replay verifier (defense-in-depth) independently re-derives the full channel transcript from serialized proof data before on-chain submission, covering all 11 layer tags (Table 2), deferred sub-proofs, IO commitment, trailing data rejection, and weight binding.

Not enforced: *LayerNorm/RMSNorm mean and variance computation (see Section 10).*

By union bound over $L = 40$ layers, the total forgery probability for the linear arithmetic trace is:

$$\varepsilon_{total} \le \underbrace{L \cdot \varepsilon_{sc}}_{sumcheck} + \underbrace{\varepsilon_{mle}}_{MLE\ opening} + \underbrace{\varepsilon_{bind}}_{binding}. \tag{11}$$

With $q = 5$: $\varepsilon_{total} < 40 \cdot 2^{-119} + 2^{-65} + 2^{-116} \approx 2^{-65}$, dominated by MLE opening.

With $q = 10$: $\varepsilon_{total} < 40 \cdot 2^{-119} + 2^{-130} + 2^{-116} \approx 2^{-111}$, dominated by binding.

Remark 8.5 (Gap between field security and system soundness). The 124-bit field QM31 provides a *per-challenge* security margin. The deployed system's end-to-end bound (2^{-111} at $q = 10$) is lower because soundness errors accumulate across L layers and M weight matrices via union bound. This gap is inherent to multi-instance interactive proof composition and is consistent with the analysis in other deployed systems [24]. Increasing M (more layers) linearly degrades the binding term; for a 100-layer model, the bound would be $\approx 2^{-110}$, still well above the 80-bit conventional threshold.

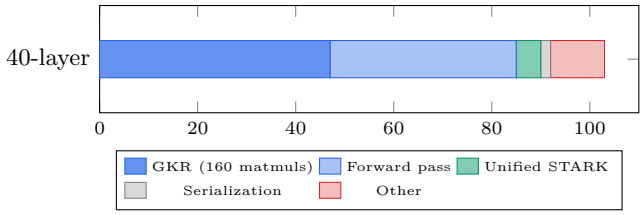

Figure 6: Full-model proving breakdown for Qwen3-14B (40 layers) on H100 NVL. GKR reductions dominate at 46%.

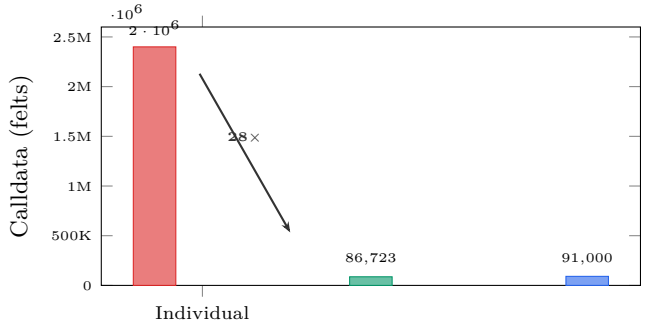

Figure 7: Calldata comparison for 40-layer Qwen3-14B. Aggregated binding is the dominant reduction (28×); increasing q from 5 to 10 adds only 4.6% more felts while improving MLE opening soundness by 65 bits.

8.6 Defense-in-Depth: Fiat-Shamir Replay Verifier

In addition to the on-chain Cairo verifier, the prover infrastructure includes a Rust-side *replay verifier* that independently re-derives the Poseidon252 Fiat-Shamir channel state from the serialized proof data. This was hardened over 10 rounds of security review:

1. **Rounds 1–3:** Infrastructure hardening (API auth, rate limiting, CORS, path traversal, secrets removal).
2. **Rounds 3–5:** RMSNorm/LayerNorm verification gaps, multi-row norm binding, SIMD flag gate.
3. **Round 6:** Deferred proof sumcheck replay and trailing data rejection.
4. **Round 7:** Double-packed channel replay; weightless deferred proof handling.
5. **Round 8:** Full tag coverage—replay handlers for all 12 GKR layer tags (Table 2); IO commitment wiring.
6. **Round 9:** Weight binding transcript replay (super-root, mismatch sumcheck, oracle eval, MLE opening).
7. **Round 10:** Piecewise activation as default; production MLE query default set to $q = 10$ (130-bit opening security); LayerNorm variance derivation binding.

The replay verifier runs before on-chain submission and rejects proofs whose serialized Fiat-Shamir transcript diverges from the prover's channel state. This catches prover/serializer mismatches, stale-binary bugs, and accidental trailing data, preventing invalid proofs from consuming gas on-chain. The replay is purely a pre-flight check; the on-chain Cairo verifier remains the authoritative trust anchor.

9 Benchmarks

9.1 Experimental Setup

9.2 Full-Model Proving

Audit pipeline: 3 multi-inference proofs across all 40 layers complete in 5 min 11 s, including forward pass replay, deterministic checking (6/6 pass), and audit report generation.

Weight cache effect: first-run weight commitment (160 Poseidon Merkle roots) takes ∼870 s on GPU; subsequent proofs hit the `.swcf` cache and reduce this to <1 ms.

Weight binding: RLC binding (default) mixes all 160 weight Merkle roots into the Fiat-Shamir channel before sumcheck challenges are drawn, providing cryptographic soundness in <1 ms. Full MLE opening proof (via `STWO_AGGREGATED_FULL_BINDING=1`) is available for trustless on-chain streaming where the verifier cannot re-check weight commitments independently.

9.3 Ablations

Table 7 and Figure 7 isolate the effect of the main design parameters. Aggregated binding is the dominant contributor to calldata reduction (28×), while increasing q from 5 to 10 adds only 4.6% calldata but improves MLE opening soundness by 65 bits. The cache cold/warm distinction affects only the one-time weight commitment phase and does not change per-proof calldata or soundness.

9.4 Decode-Step Proving

Table 8 shows the per-step cost of decode proving. Weight commitment caching (Section 3.3) eliminates 0.8 s per step after the first cold pass, yielding a 19% reduction in steady-state latency.

The incremental Merkle KV commitment reduces per-step Poseidon hashes from $O(\text{cache_size})$ to $O(\text{new_tokens} \times \log \text{capacity})$. At $T = 4096$ with 40 layers and 8 KV heads, the flat scheme requires ∼524K Poseidon hashes per step, while the incremental tree requires only ∼140—a 3,700× reduction. Although the absolute time saving is modest at current cache sizes (∼50 ms → <1 ms), the asymptotic improvement is critical for long-context inference where T may reach 32K–128K.

9.5 Comparison with Prior Work

Table 9 positions ObelyZK relative to existing systems. Entries marked ‡ reflect systems for which specific benchmarks or parameter counts were not available in public documentation as of March 2026; we report only what is publicly verifiable and encourage the respective teams to publish comparable measurements. Key differentiators:

- **Scale + deployment**: to our knowledge, the first publicly described system combining 14 B-scale linear-trace proving with a deployed on-chain Cairo verifier on Starknet.

- **Aggregated binding**: enables practical calldata for LLM-scale on-chain verification; we are not aware of a comparable batched-opening mechanism in other ZKML systems.
- **Honest scope**: ObelyZK constrains only linear operations; zkLLM constrains full inference including non-arithmetic ops via *tlookup*.

9.6 On-Chain Costs

9.7 Codebase Metrics

10 Limitations and Future Work

10.1 Activation Verification via Range-Reduced LogUp

As of v34, nonlinear activation functions (GELU, Sigmoid, Softmax) are constrained via **range-reduced LogUp**, and ReLU is verified algebraically (product+binary eq-sumcheck).

Range reduction. M31 matmul intermediate outputs span the full field ($2^{31}-1$) but precomputed activation tables have 2^{16}–2^{20} entries. We apply the same range-masking pattern used by LayerNorm rsqrt verification: mask each activation input to the table range ($\texttt{val}\ \&\ ((1 \ll \texttt{log_size}) - 1)$), then run LogUp proving that (masked_in, f(masked_in)) is in the precomputed table.

Soundness model. This verifies activation correctness on the lower 16–20 bits. The attack surface is narrowed from "arbitrary function substitution" to "high-bit-only corruption," which is infeasible in fixed-point arithmetic where the low bits carry the signal (quantized model weights and activations are concentrated in the lower bits).

Calldata overhead. Per non-ReLU activation layer: ∼336 felts (184 for LogUp eq-sumcheck + 152 for multiplicity sumcheck). For Qwen3-14B (80 GELU layers): ∼26,880 extra felts, a ∼31% increase over the 86,723-felt linear-only baseline.

Enforcement. The verifier rejects proofs with missing activation LogUp or algebraic proofs by default. Legacy proofs without activation verification can be accepted by setting `STWO_ALLOW_MISSING_ACTIVATION_PROOF=1` (opt-out, not opt-in).

10.2 Remaining Gap: LayerNorm/RMSNorm Statistics

LayerNorm rsqrt values are verified via LogUp, and the linear scaling output = (input − μ) × rsqrt is proven via eq-sumcheck. However, the **mean** (μ) and **variance** computation are not independently constrained—the prover can claim arbitrary μ and σ^2 values. A future inner-product sumcheck for mean and degree-3 eq-sumcheck for variance will close this gap.

10.3 End-to-End Multi-Layer Timing

The full 40-layer Qwen3-14B inference (9.2 B parameters, 281 layers, 160 MatMul + 81 RMSNorm + 40 GELU) has been proved end-to-end on a single H100 NVL GPU in **103 s** (2.58 s amortized per transformer layer). This includes forward pass, GKR sumcheck over all 160 matmul reductions, RLC weight binding via Fiat-Shamir channel-mixed Poseidon commitments, unified STARK for non-matmul components, and serialization. Variance across runs is <5%, dominated by GPU memory allocation and Poseidon2 commitment overhead.

The multi-inference audit pipeline proves 3 full inferences across all 40 layers—480 matmul proofs with deterministic checking and audit report generation—in 5 min 11 s (5m 11s).

10.4 Query Count

As of v34, the production default is $q = 10$ MLE queries, yielding 130-bit opening security (matching the 124-bit field security target). The previous default ($q = 5$, 65-bit) is available via `MLE_NUM_QUERIES=5` for testing. This is a deployment parameter; no protocol changes were required.

10.5 Security Hardening (v34)

The following security measures were added and verified via a 41-test audit-grade integration suite:

DAG residual weight binding. Models with residual connections (skip/Add layers) create deferred proofs for the non-trunk branch. In `AggregatedOracleSumcheck` mode, deferred MatMul weight claims must be included in the aggregated binding—without this, the prover's claim count mismatches the verifier's reconstruction, enabling weight fabrication attacks on DAG architectures. All five prover variants and both verifier paths (CPU, SIMD) now consistently include deferred weight claims.

Grouped binding for scalability. Models with > 4 weight matrices split the aggregated binding into per-layer groups of 4 matrices each. Each group produces a 2^{30}-element virtual MLE (∼17 GB) that fits in GPU VRAM, instead of a single 2^{36+}-element MLE (> 1 TB) that is infeasible. Groups are proved sequentially through the same Fiat-Shamir channel, maintaining transcript soundness.

Tamper detection coverage. 18 end-to-end adversarial tests cover: tampered round polynomials (single/deep-chain), forged weight claims (value, eval point, node ID), wrong activation types, corrupted LogUp multiplicities, tampered final evaluations (A/B), deferred proof round poly and weight claim forgery, and tampered input/output MLEs. All tamper vectors are detected and rejected.

Security gate enforcement. 23 integration tests verify that default soundness gates reject: missing piecewise activation proofs, missing norm variance proofs, missing segment binding, RLC-only binding in streaming mode, and tampered streaming calldata batches.

10.6 Future Directions

1. **LayerNorm mean/variance consistency**: add inner-product sumcheck for mean + degree-3 eq-sumcheck for variance, closing the remaining normalization gap.

2. **Recursive STARK compression**: compose streaming proofs into a single sub-1 KB recursive proof, enabling constant-size on-chain verification.
3. **Verifiable fine-tuning**: extend GKR to backward pass gradients.
4. **Long-context decode**: stress-test decode chain registry at $T = 32\text{K}$–128K sequences, optimizing incremental Merkle capacity doubling.

11 Related Work

ZKML systems. EZKL [6] handles ∼1 M-parameter models via Halo2/KZG with Solidity verifiers. Giza [8] uses STARK (Stone) for small–medium models. Polyhedra's Expander [19] is a GKR-based engine achieving 4,500 Keccak-f/s. Lagrange DeepProve [14] targets large-scale GKR with GPT-2 demonstrations. zkLLM [22] (ACM CCS 2024) introduces *tlookup* for non-arithmetic tensor ops, proving up to 13 B parameters in 1–15 minutes with <200 KB proofs. A comprehensive survey appears in [25]. zkCNN [16] applies GKR to convolutional networks. vCNN [15] uses zkSNARKs for CNN verification.

Interactive proofs. The GKR protocol [9] was made practical by Thaler [23] and extended by Cormode et al. [4]. The sumcheck protocol [17] underpins all GKR reductions. Schwartz-Zippel [20, 26] provides the foundational polynomial identity testing bound. Thaler's textbook [24] provides a thorough treatment of interactive proofs and their applications.

Proof infrastructure. STWO [21] provides the Circle STARK backend [12] with M31-native arithmetic. FRI [1] enables efficient proximity testing for STARK proof composition. Marlin [3] offers preprocessing zkSNARKs with universal SRS as an alternative to STARKs.

12 Conclusion

We presented ObelyZK, a system for verifiable linear-trace inference at LLM scale. The core contribution—aggregated weight binding—reduces on-chain calldata by 28× (from 2.4 M to 86,723 felts), making streaming verification of 40-layer transformer proofs practical within Starknet's transaction limits. The system achieves 103 s end-to-end GPU proving for the full 40-layer model (2.58 s amortized per layer) and is deployed on Starknet Sepolia with a 14 K-line Cairo verifier.

The deployed verifier (v34) constrains the linear arithmetic trace (matmul, addition, scaling), nonlinear activations (GELU, Sigmoid, Softmax via range-reduced LogUp; ReLU algebraically), and KV-cache commitment chains with on-chain decode chain enforcement—RoPE position offset binding, cross-proof continuity assertions, and enriched verification events for external auditing. Incremental Poseidon Merkle trees reduce per-step commitment cost from $O(T)$ to $O(\log T)$, and weight commitment caching eliminates redundant Merkle root computation across decode steps. A 10-round-hardened Fiat-Shamir replay verifier provides defense-in-depth, independently checking the full channel transcript—all 12 layer tags, deferred proofs, IO commitment, and weight binding—before on-chain submission, with $q = 10$ MLE queries providing 130-bit opening security. The remaining gap is LayerNorm/RMSNorm mean/variance computation; with norm statistics verification the system would achieve full inference verification.

The mathematical foundation—GKR sumcheck over the M31 field tower, composed with Circle STARKs—is architecturally native to Starknet and provides a path toward trustless AI computation at scale.

A Poseidon2-M31 Specification

Poseidon2-M31 [10] is a sponge-based hash operating natively over M31. Parameters: state width $t = 16$, rate=capacity= 8, S-box degree $d = 5$, full rounds $R_f = 8$ (4+4 split), partial rounds $R_p = 14$, total 22 rounds.

S-box validity: $\gcd(5, p-1) = 1$, confirming $x \mapsto x^5$ is a permutation. Inverse: $d^{-1} = 5^{-1} \bmod (p-1) = 1{,}717{,}986{,}917$.

Diffusion. External: circulant MDS $\mathrm{circ}(2M_4, M_4, M_4, M_4)$. Internal: $M_I = J + \mathrm{diag}(\mathbf{v})$, $v_0 = p - 2$.

Hash output: 8 M31 elements (248 bits, ∼124-bit collision). Compression (16 → 8): domain constant `0x4D524B4C` ("MRKL").

B Privacy Protocol Summary

ObelyZK includes a privacy protocol built on Poseidon2-M31 with six DeFi mechanisms (Table 12). This is architecturally independent of the ZKML verification system and is described here for completeness.

Note structure: Commitment (11 M31 inputs): $\mathrm{Poseidon2}(\mathrm{pk}[8], \mathrm{blinding}, \mathrm{amt_lo}, \mathrm{amt_hi})$. Nullifier (12 M31 inputs): $\mathrm{Poseidon2}(\mathrm{comm}[8], \mathrm{sk}[4])$. Transaction circuits: deposit (2 Poseidon2 permutations), spend (2-in/2-out with depth-20 Merkle membership), withdraw (1-in with withdrawal binding).

C Deployed Contract Addresses

D Notation Reference

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Table 2: GKR layer types and reduction strategies. All 12 layer tags are constrained in the deployed verifier (v34).

Layer	**Tag**	**Deg.**	**Rnds**	**Strategy**
MatMul(m, k, n)	0	2	$\log_2 k$	Product of two MLEs
Add(n)	1	1	1	Linearity: $c_0 =$ claim/2
Mul(n)	2	2	1	Degree-2 check
Activation	3	3	$\log_2 n$	LogUp (GELU/Sigmoid/Softma or algebraic (ReLU)
LayerNorm	4	3	$\log_2 n$	Linear eq-sum + rsqrt LogUp
GQA Attention	5	2	decomposed	Q×K, ×V sub-proofs
Dequantize	6	3	$\log_2 n$	Range-reduced LogUp
MatMulDualSimd	7	3	$\log_2 k$	Block-partitioned degree-3
RMSNorm	8	3	$\log_2 n$	Linear eq-sum + rsqrt LogUp
Quantize	9	3	$\log_2 n$	Range-reduced LogUp
Embedding	10	3	$\log_2 n$	Sparse table LogUp
AttentionDecode	11	2	decomposed	Cached KV decode attention

Table 3: Calldata for Qwen3-14B 40-layer proof. Aggregated binding reduces total calldata by 28×.

Method	**Felts**	**Bytes**
Individual openings	∼2,400,000	∼76.8 MB
Aggregated binding	86,723	∼2.8 MB
GKR layer data	∼82,433	∼2.6 MB
Binding proof	∼4,290	∼137 KB
Reduction	**≈28×**	

Table 4: Key Cairo verifier modules.

Module	**LOC**
`aggregated_binding.cairo`	995
`gkr.cairo`	547
`field.cairo`	503
`types.cairo`	367
`sumcheck.cairo`	269
`mle.cairo`	250+
`channel.cairo`	244

Table 5: Benchmark configuration.

GPU	NVIDIA H100 80 GB SXM
CPU	AMD EPYC 7763 (128 cores)
Model	Qwen3-14B (HuggingFace checkpoint)
Parameters	14.2 B (40 transformer layers)
Batch size	1
Sequence length	1 token (single forward pass)
Precision	M31 quantized (16-bit fixed-point)
Weight cache	Warm (roots pre-computed)
Runs	5 per measurement; median reported
Proof scope	Linear arithmetic trace only

Table 6: Qwen3-14B full 40-layer proving breakdown (H100 NVL, cached weights). *Measured* end-to-end timings for the complete 9.2 B-parameter model.

Phase	**Time**
Model loading (160 weights)	35 s
Forward pass (281 nodes)	38 s
GKR layer reductions (160 matmuls)	47 s
Weight binding (RLC, Fiat-Shamir)	<0.01 s
Unified STARK (121 components)	5 s
Serialization	2 s
Total (40 layers)	**103 s**
Amortized per layer	**2.58 s**

Table 7: Ablation study: effect of key parameters on calldata and soundness. All rows assume 40 layers, 160 matrices, $n = 13$ folds.

Configuration	**Felts**	$\varepsilon_{\mathbf{mle}}$	$\varepsilon_{\mathbf{total}}$
Individual openings	2.4 M	2^{-65}	2^{-65}
Aggregated, $q=5$	86,723	2^{-65}	2^{-65}
Aggregated, $q=10$	∼91 K	2^{-130}	2^{-111}
Cold cache, $q=5$	86,723	2^{-65}	2^{-65}
Timing impact			
Warm cache	<1 ms weight commit		
Cold cache	∼500 s weight commit (first run)		
$q=5 \to q=10$	+4,000 felts (+4.6%), no proving time change		

Table 8: Decode-step proving cost breakdown (single H100, Qwen3-14B 40-layer, prefill $T=64$). Weight-cache column shows subsequent-step cost after the initial cold pass populates the cache.

Component	**Cold**	**Cached**
Forward pass (1 token)	0.4 s	0.4 s
GKR proof (40 layers)	2.8 s	2.8 s
Weight commitments	0.8 s	<1 ms
KV commitment (flat rehash)	50 ms	50 ms
KV commitment (incr. Merkle)	<1 ms	<1 ms
Overhead (channel, serde)	0.3 s	0.3 s
Total (flat rehash)	**4.4 s**	**3.6 s**
Total (incr. Merkle)	**4.3 s**	**3.5 s**

Table 9: Comparison with existing ZKML systems. †Linear trace only (activations unconstrained). ‡Estimated from published data. ⋆Measured end-to-end (40 layers on single H100).

System	Max Params	Prove Time	Full Semantics	Deployed Verifier	Proof Size	Measured	Proof System
EZKL [6]	∼1 M	seconds	✓	Solidity	<1 KB	✓	Halo2/KZG
Giza [8]	∼10 M	minutes	✓	Cairo (Stone)	KB–MB	✓	STARK
zkLLM [22]	13 B	1–15 min	✓	None	<200 KB	✓	Custom
Expander [19]	—‡	—‡	—‡	None	—‡	—	GKR
DeepProve [14]	GPT-2‡	—‡	—‡	None	—‡	—	GKR
ObelyZK	14 B†	**103 s⋆**	✗†	Cairo v34	7.1 MB	✓	GKR+STARK

Table 10: On-chain verification costs (Starknet Sepolia).

Operation	Gas	Cost
GKR verify (1 layer)	∼280 K	<$0.001
Full 40-layer (18 TXs)	∼2.5 M	∼$0.01

Table 11: Codebase scale (`stwo-ml` repository).

Component	Lines	Lang.
`stwo-ml` prover	117,000+	Rust
CUDA kernels	20,000+	CUDA C++
Cairo verifier	∼14,000	Cairo
Library tests passing	1,029+	Rust
E2E audit tests	10	Rust
Cairo verifier tests	95	Cairo

Table 12: Privacy mechanisms (deployed on Starknet Sepolia).

Mechanism	Basis
Privacy Pools (ASP [2])	Poseidon2-M31, LeanIMT
Dark Pool Auctions	Pedersen commit-reveal
Shielded AMM Swaps	ZK withdrawal proofs
Stealth Payments	ECDH + Schnorr (EIP-5564 [18])
UTXO Privacy Pool	Poseidon2-M31, depth-20 Merkle
Confidential Transfers	ElGamal [5] on Stark curve

Contract	Starknet Sepolia
ZKML Verifier	`0x005928ac548dc2719ef1...674fba`
GKR v31 (class)	`0x6a6b7a75d5ec1f63d715...5b5439`
Deployer	`0x0759a4374389b0e3cfcc...bb344`

Symbol	Meaning
M31	Mersenne prime field $\mathbb{F}_{2^{31}-1}$
CM31	Complex extension $\mathrm{M31}[i]/(i^2+1)$
QM31	Quartic extension $\mathrm{CM31}[j]/(j^2-(2+i))$
$\widetilde{f}$	Multilinear extension of f
$\mathrm{eq}(\mathbf{x},\mathbf{y})$	Lagrange equality kernel
$\rho,\alpha,\lambda,\beta,\gamma$	Fiat-Shamir challenges
$\mathcal{P},\mathcal{V}$	Prover, Verifier
ε	Soundness error bound
q	MLE spot-check query count per fold
ASP	Association Set Provider
GKR	Goldwasser-Kalai-Rothblum protocol
LogUp	Logarithmic derivative lookup argument